

ScanScript Guide

How to write pattern scripts for full ASX universe scans · ASX Sector Intelligence

ScanScript is the desk's text language for defining *My Patterns*. You write simple rules in a text box; the app compiles them into scanner conditions and runs them across every ASX stock — no TradingView Pine, no webhooks, no code execution.

1. Quick start — create a pattern in the app

- 1 Open any stock chart (or the main desk) and open the **Pattern Analysis** panel on the right.
- 2 Under **Create my pattern**, click **New**.
- 3 Enter a **pattern name**, choose **Bullish / Bearish / Neutral**, and optional notes.
- 4 Under **How to detect**, select **ScanScript (text rules)**.
- 5 Type your script in the text area. Red errors appear if syntax is wrong; a green *Compiles to...* line shows the parsed rules.
- 6 Click **Save private pattern**. The scanner runs on charts and on the full industry scan.

2. Script structure

A script is plain text, one statement per line. Empty lines and lines starting with `#` are ignored.

```
# Oversold with volume – example header comments
bias bullish
match all

rsi(14) <= 35
rvol >= 1.5
above_sma(50)
```

Optional headers

Line	Meaning
<code>match all</code>	Every condition must be true (AND). Default if omitted.
<code>match any</code>	At least one condition must be true (OR).
<code>bias bullish</code>	Sets pattern bias to bullish (overrides the form dropdown).
<code>bias bearish</code>	Sets pattern bias to bearish.
<code>bias neutral</code>	Sets pattern bias to neutral.

Headers accept an optional colon: `match: all` and `bias: bullish` also work.

3. Conditions reference

Each condition line compares a metric to a number using `>`, `>=`, `<`, `<=` (Unicode `≥` / `≤` also accepted).

RSI

```
rsi(14) <= 30
rsi(14) >= 70
```

14-period RSI on daily closes. Range 0–100. Only period **14** is supported. Aliases: `rsi14`.

Relative volume (RVOL)

```
rvol >= 1.5
relative_volume >= 2
```

Today's volume divided by the average of the previous 20 sessions. `1.5` = 50% above average. Aliases: `relative_volume`, `rel_volume`.

Percent above moving averages

```
above_sma(50)
below_sma(200)
above_ema(21)
below_ema(21)
```

Shorthand: `above_sma(50)` means close is above the 50-day SMA ($\geq 0\%$ above). `below_sma(50)` means below ($< 0\%$).

With explicit threshold:

```
above_sma(50) >= 2    # at least 2% above 50 SMA
below_sma(20) <= -3    # at least 3% below 20 SMA
```

Function	Periods	Meaning
above_sma(n) / below_sma(n)	20, 50, 200	% distance of close from SMA
above_ema(n) / below_ema(n)	21 only	% distance of close from EMA

Negative values mean price is below the average (e.g. `-2` = 2% below).

Percent change

```
pct_chg(5) >= 5
pct_chg(20) >= 10
pct_change(5d) >= 3
```

Percentage change over 5 or 20 trading days. Aliases: `pct_change` , `change` . Optional `d` suffix on the argument (`5d`).

4. Limits and behaviour

Topic	Detail
Max conditions	8 per script (form UI allows 4; ScanScript allows more).
Lookback	A hit fires if conditions are true on any of the last ~10 daily bars within your scan window.
Scan window	Use Pattern panel buttons (e.g. 1M, 3M, 6M) — only hits whose start date falls in that window count.
Data	Daily OHLC from Yahoo. Stocks without data are skipped (same as other scanners).
Security	Scripts compile to fixed metrics only — no loops, functions, or arbitrary code.

5. Example scripts

Oversold bounce (bullish)

```
bias bullish
match all

rsi(14) <= 35
rvol >= 1.5
above_sma(200)
```

RSI oversold, elevated volume, still above long-term trend.

Stage-style trend (bullish)

```
match all

above_sma(50)
above_sma(200)
pct_chg(20) >= 8
rsi(14) >= 45
rsi(14) <= 75
```

Above 50 and 200 SMA, 20-day momentum, RSI not extended.

Volume breakout

```
match all

rvol >= 2
pct_chg(5) >= 3
above_sma(20)
```

Bearish weakness

```
bias bearish
match all

rsi(14) >= 70
below_sma(50)
pct_chg(5) <= -3
```

Flexible filter (OR)

```
match any

rsi(14) <= 30
rsi(14) >= 70
```

Flags either oversold or overbought extremes.

6. Where results appear

- **Chart modal** — Pattern Analysis → **My Patterns** category shows hits for the open stock.
- **Sector Table** — My Pattern hits appear as chips on each stock row (labelled with your pattern name).
- **Industry scan** — Background scan evaluates all tickers when OHLC is loaded; results are cached locally.

My Patterns are **private** (stored in your browser). They are separate from **Special Patterns** (♦) and starred chart patterns (★).

7. Troubleshooting

Error / issue	Fix
invalid syntax	Check operators and parentheses. One condition per line.
Add at least one condition	Script has only headers/comments — add metric lines.
max 8 conditions	Split into two patterns or simplify rules.
pct_chg supports 5 or 20	Use pct_chg(5) or pct_chg(20) only.
No hits on desk	Widen scan window; confirm Yahoo data exists for tickers; conditions may be too strict.
Save button disabled	Fix red validation errors in the script editor first.

8. ScanScript vs Pine Script

If you know **TradingView Pine Script**, ScanScript will feel familiar in spirit: you describe indicator filters with comparisons, combine them with AND/OR, and look for setups on OHLC bars. The big difference is *purpose*: Pine builds chart indicators and strategies on TradingView; ScanScript builds a **safe desk scanner** that runs across the ASX universe inside this app.

Mental model: think of ScanScript as a short Pine-style condition list — not a full programming language. There are no plots, loops, arrays, `strategy.*`, webhooks, or arbitrary functions. Each line compiles to one fixed desk metric.

What is the same (conceptually)

Idea in Pine	ScanScript equivalent
<code>rsi(close, 14) <= 35</code>	<code>rsi(14) <= 35</code>
<code>ta.sma(close, 50) / price</code> above MA	<code>above_sma(50)</code> or <code>above_sma(50) >= 2</code>
Volume vs average volume	<code>rvol >= 1.5</code>
AND / OR filters	<code>match all</code> / <code>match any</code>
Comments with <code>//</code>	Comments with <code>#</code>

Side-by-side example

Pine-style thinking (not runnable Pine here):

```
// Oversold with volume, still in uptrend
rsi0k = ta.rsi(close, 14) <= 35
vol0k = volume > ta.sma(volume, 20) * 1.5
trend0k = close > ta.sma(close, 50)
signal = rsi0k and vol0k and trend0k
```

Same idea as ScanScript:

```
bias bullish
match all

rsi(14) <= 35
rvol >= 1.5
above_sma(50)
```

What is different

Pine Script	ScanScript
Full language (vars, loops, plots, strategies)	Declarative condition lines only
Runs on TradingView charts / alerts	Runs on this desk's ASX OHLC scan
You can invent custom series math	Only supported metrics (RSI, RVOL, SMA/EMA %, % change)
Can execute arbitrary script logic	Compiles to fixed safe rules — no code execution
Often one symbol / alert feed	Saved pattern scans the full ASX universe

Tip: when translating from Pine, keep only the boolean filter lines (`rsi ...`, `close > sma ...`, volume spike). Drop plots, inputs, colors, and strategy entries — those do not exist in ScanScript.

9. ScanScript vs other pattern types in this app

Method	Best for
ScanScript	Multi-condition RSI / volume / MA / momentum filters across full ASX.
My conditions (form)	Same metrics, up to 4 rules, no typing.
Candle shape builder	Hammer, engulfing-style geometry on daily/weekly candles.
Reuse built-in scanner	Alias an existing catalog pattern under your own name.
Special Patterns (♦)	Desk-maintained weekly/snapshot rules — not user-editable.

TraderScope · ScanScript v1 · Pattern authoring guide (incl. Pine Script comparison)
For the latest metrics and UI, refer to Create pattern → ScanScript in the app.